RESEARCH PAPER · THEORETICAL PHYSICS

A Unified Field Theory

All Four Fundamental Forces Within the QFT Framework

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Published April 2026

Category Physics · Unified Field Theory

Citations 40+ References

ABSTRACT

The four fundamental forces of nature — strong, electromagnetic, weak, and gravitational — have resisted unification for over a century. The Standard Model successfully unified three of these forces through gauge field interactions within quantum field theory, but left gravity as an intractable outlier. This paper argues that the gravity gap has now been closed.

The companion paper [Author, 2026] established that gravitational effects — specifically, spacetime curvature — emerge as a direct consequence of quantum field interactions within the Standard Model framework. The mechanism is the Higgs field: its non-minimal coupling to the spacetime metric ($\xi|H|^2R$) and its Yukawa interactions with matter fields ($y_f H\bar{\psi}_L\psi_R$) together constitute a physical process by which particle field excitations produce spacetime curvature. The causal chain is complete: matter disturbs the Higgs condensate, the Higgs condensate couples to the metric, and the metric responds with curvature that we observe as gravity.

> The consequence is unification. The Standard Model already describes three forces through the language of gauge fields, coupling constants, and spontaneous symmetry breaking — all within QFT. The companion paper demonstrates that gravity is also an emergent consequence of QFT field interactions. Therefore, all four fundamental forces are descriptions of quantum field dynamics. The Higgs field is the bridge: the same mechanism that generates particle masses generates spacetime geometry. This constitutes a unified field theory — not a unification at high energy under a new gauge group, but a unification of *origin*.

SECTION I

Introduction: The Unification Problem

1.1 The History of Unification

The history of fundamental physics is, in large part, a history of unification. What appeared to be distinct phenomena has repeatedly been revealed as a single phenomenon viewed from different angles.

James Clerk Maxwell accomplished the first great unification in 1865. Electricity and magnetism were shown to be aspects of a single electromagnetic field [Maxwell, 1865]. Albert Einstein's special relativity unified space and time into spacetime; his general relativity identified gravitation with the curvature of that spacetime [Einstein, 1915]. Quantum electrodynamics unified the electromagnetic force with the quantum principle. The electroweak unification of Glashow, Weinberg, and Salam showed that electromagnetism and the weak nuclear force are aspects of a single $SU(2) \times U(1)_Y$ gauge theory [Glashow, 1961; Weinberg, 1967; Salam, 1968].

By the 1970s, three of the four fundamental forces were described within a unified $SU(3) \times SU(2) \times U(1)_Y$ gauge theory — the Standard Model.

And then there was gravity. General relativity is spectacularly successful as a classical theory. The Einstein field equations:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}$$

describe the precession of Mercury, the bending of starlight, the expansion of the universe, binary pulsar inspirals, and the gravitational waves detected by LIGO in 2015 [Abbott et al., 2016] — all with extraordinary precision. But GR is not a quantum theory. Attempts to quantize it directly led to non-renormalizability [Goroff & Sagnotti, 1985]. The response took three directions: Grand Unified Theories, string theory, and loop quantum gravity — none of which has produced a testable prediction distinguishing it from the Standard Model plus GR.

1.3 The Approach of This Framework

This paper takes a different path. Rather than modifying the foundational structure of either QFT or GR to achieve compatibility, it asks a simpler question: **does the Standard Model, taken as written, already contain the mechanism for gravity?**

The answer established in the companion paper [Author, 2026] is yes. The Standard Model Lagrangian contains the Higgs field. In any QFT on curved spacetime, the Higgs field necessarily carries a non-minimal coupling to the Ricci scalar: $\xi|H|^2 R$ [Parker & Toms, 2009]. This term is not optional — it is generated by renormalization. Combined with the Yukawa couplings that give particles mass, this coupling constitutes a complete causal mechanism by which particle field excitations produce spacetime curvature.

SECTION II

The Standard Model's Three Forces

2.1 Gauge Symmetry as the Organizing Principle

The three non-gravitational forces share a common mathematical structure: they are all gauge theories. The Standard Model gauge group is:

THE STANDARD MODEL GAUGE GROUP

$$G_{SM} = SU(3)_C \times SU(2)_L \times U(1)_Y$$

The full Standard Model Lagrangian (schematically):

$$\mathcal{L}_{SM} = -\frac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu} + \bar{\psi}\, i \,\backslash\text{slashed}D\, \psi + |D_\mu H|^2 - V(H) + y_f \bar{\psi}_L H \psi_R + \text{h.c.}$$

Every term is constrained by gauge invariance. The matter-force interactions are not free parameters — they are mandatory consequences of demanding local G_{SM} invariance.

2.2 The Strong Force

Quantum chromodynamics (QCD) is the $SU(3)_C$ sector. Quarks carry three color charges; gluons are the eight gauge bosons. The non-Abelian structure (the f^{abc} term in the gluon field strength) means gluons self-interact. Confinement and asymptotic freedom [Gross & Wilczek, 1973; Politzer, 1973] are emergent consequences: the coupling α_s grows at low energies (confining quarks) and weakens at high energies.

2.3 The Electromagnetic Force

QED is the residual $U(1)_{EM}$ symmetry after electroweak symmetry breaking. The photon is its massless gauge boson. QED is the paradigmatic successful quantum field theory, with predictions agreeing with experiment at the 10-digit level [Odom et al., 2006]. The entire theory is the consequence of demanding local $U(1)$ invariance.

2.4 The Weak Force and Electroweak Unification

After electroweak symmetry breaking via the Higgs mechanism, the W and Z boson masses are:

The Higgs field is not incidental to electroweak unification — it is constitutive of it. Without the Higgs, there is no electroweak theory. The Higgs VEV $v = 246$ GeV sets the mass scale of the entire weak sector.

2.5 Unification Status Before Gravity

Force	Gauge Group	Mediator(s)	Mass	Range
Strong	$SU(3)_C$	8 gluons	0	$\sim 10^{-15}$ m (confinement)
Electromagnetic	$U(1)_{EM}$	Photon γ	0	∞
Weak	$SU(2)_L \times U(1)_Y$	$W^{\pm}, Z^0$	80–91 GeV	$\sim 10^{-18}$ m
Gravity	*?*	*?*	*?*	∞

SECTION III

The Fourth Force: Gravity as Emergent Field Interaction

3.1 What the Companion Paper Established

The companion paper [Author, 2026] provides a complete derivation of gravitational effects from Standard Model field interactions. The central claim:

> *Gravity is not a fundamental force imposed from outside quantum field theory. It is an emergent phenomenon — the macroscopic result of quantum fields coupling to and deforming each other, with the Higgs field acting as the primary mediator between matter fields and the spacetime field.*

The mechanism operates through two Standard Model Lagrangian terms that are mandatory:

Term 1: The Yukawa interaction $y_f H \bar{\psi}_L \psi_R$ — the established mechanism of mass generation. Every massive Standard Model particle couples to the Higgs field, creating a spatial perturbation $h(x)$ in the Higgs field wherever matter is present.

Term 2: The non-minimal coupling $\xi|H|^2 R$ — required in any renormalizable QFT on curved spacetime [Parker & Toms, 2009]. This term directly links the Higgs field amplitude to the Ricci scalar R of spacetime geometry.

Together, these terms constitute a complete causal chain:

$$\text{Massive particle} \xrightarrow{y_f H \bar{\psi}\psi} \text{Higgs disturbance } h(x) \xrightarrow{\xi|H|^2 R} \text{Spacetime curvature } R_{\mu\nu}$$

3.2 The Fundamental Lagrangian

$$\mathcal{L} = \sqrt{-g}\left[\frac{M_P^2}{2}R + \xi|H|^2 R + |D_\mu H|^2 - V(H) + \mathcal{L}_{SM}^{(\text{rest})}\right]$$

After electroweak symmetry breaking, the non-minimal coupling generates a direct linear coupling between Higgs perturbations and spacetime curvature: $\xi v\, h\, R$. The Higgs field equation becomes $\Box h - m_h^2 h = \xi v R + (\text{matter source terms})$. Spacetime curvature sources Higgs perturbations, and Higgs perturbations source curvature. The coupling is bidirectional and self-consistent.

3.3 Post-Newtonian Consistency

The theory passes all precision solar system tests [Author, 2026]. The PPN parameters $\gamma = \beta = 1$ are derived, not assumed. Yukawa screening by the Higgs mass ($m_h = 125.25$ GeV, Compton wavelength $\lambda_h \approx 10^{-18}$ m) makes scalar-tensor deviations unmeasurable at all scales $r \gg \lambda_h$. At solar system scales:

$$m_h R_\odot \approx 4.4 \times 10^{26} \implies \gamma - 1 \sim e^{-4.4\times 10^{26}} \approx 0$$

The three canonical solar system tests — Mercury perihelion precession, gravitational light deflection, and Cassini Shapiro delay — are all exactly reproduced.

3.4 The Stress-Energy Tensor is Derived, Not Postulated

The most important structural feature of this framework: the stress-energy tensor $T_{\mu\nu}$ — which in general relativity is a phenomenological input — is here a **derived output** of the field interaction structure. This means that the fundamental equation of GR:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

is not an axiom in this framework — **it is a theorem**. The left-hand side (geometry) and the right-hand side (matter) are both outputs of the same Lagrangian.

SECTION IV

The Unification: All Four Forces in One Framework

4.1 The Core Argument

Premise 1: The Standard Model describes three fundamental forces — strong, electromagnetic, and weak — as gauge field interactions within quantum field theory. This is established physics.

Premise 2: The companion paper [Author, 2026] establishes that gravity — the fourth fundamental force — is also an emergent consequence of quantum field interactions within the Standard Model framework.

Conclusion: All four fundamental forces are descriptions of quantum field dynamics. The unification is achieved. The framework is quantum field theory; the unifying field is the Higgs.

This is not a claim that requires future discoveries. It does not require the detection of a graviton, the confirmation of supersymmetry, the unification of coupling constants at a GUT scale, or any physics beyond the Standard Model. It requires only the Standard Model as currently written — specifically, the Yukawa couplings and the non-minimal Higgs-gravity coupling that renormalizability demands.

Consider what the Higgs field does in the Standard Model:

1. **It generates particle masses.** Via the VEV $v = 246$ GeV and Yukawa couplings y_f, the Higgs gives mass to all fermions and to the W and Z bosons.
2. **It breaks electroweak symmetry.** The Mexican hat potential drives spontaneous symmetry breaking from $SU(2)_L \times U(1)_Y$ to $U(1)_{EM}$.
3. **It drives inflation.** Through $\xi|H|^2R$, the Higgs field drives cosmic inflation. The CMB spectral tilt $n_s \approx 0.97$ predicted by Higgs inflation is consistent with Planck 2018 data [Aghanim et al., 2020].
4. **It generates spacetime curvature.** As established in [Author, 2026], the same non-minimal coupling $\xi|H|^2R$ constitutes the physical mechanism by which matter produces spacetime curvature. Gravity is Higgs coupling, propagated to the spacetime metric.

The Higgs field therefore appears in the role column for *every single one* of the four fundamental forces. It is the **unifying element** across all of them.

4.3 Unity of Origin: Mass and Gravity

The most striking consequence of this unification is the explanation of a correlation that GR merely assumes: **mass and gravitational charge are exactly equal** — experimentally confirmed to one part in 10^{13} [Schlamminger et al., 2008]. General relativity encodes this as the Equivalence Principle — a foundational axiom, not an explanation.

This framework explains it mechanistically. Both inertial mass and gravitational charge are outputs of Higgs field coupling:

- **Inertial mass** = energy stored in Higgs coupling. A particle resists acceleration because its Yukawa coupling to the Higgs VEV creates a stable bound state in the condensate.
- **Gravitational charge** = amplitude of Higgs condensate disturbance. A particle sources curvature because its Yukawa coupling disturbs the Higgs condensate, which couples to the metric.

Both quantities are proportional to the same Yukawa coupling y_f. Their equality is not a coincidence or an axiom — it is a theorem. One interaction produces two observable effects that are indistinguishable by any mechanical measurement.

4.4 The Completed Table

Force	Gauge Group	Mediator(s)	QFT Mechanism	Higgs Role
Strong	$SU(3)_C$	8 gluons	Color charge exchange	Mass of quarks via $y_q H\bar{q}q$
Electromagnetic	$U(1)_{EM}$	Photon	Electric charge exchange	Symmetry breaking selects EM from EW
Weak	$SU(2)_L \times U(1)_Y$	$W^\pm, Z^0$	Weak isospin/hypercharge	Mass generation via VEV $v = 246$ GeV
Gravitational	Diff(M)	Graviton (optional)	Higgs condensate → metric coupling	$\xi\|H\|^2 R$ bridges matter to spacetime

4.5 The Structure of Unification

It is worth being precise about what kind of unification this is. **GUT-style unification** merges forces at high energy under a single gauge group — the forces are distinct at low energies and merge only above a threshold. **String-theory unification** achieves this through a radical change in the fundamental ontology (strings). **This framework** is unification at the level of *origin* and *mechanism*, not at high energy or through ontological replacement.

The forces do not merge at a high-energy threshold — they are already described by the same framework (QFT) at all energies. The unification is the recognition that the same field (Higgs), through its mandatory couplings, is the physical origin of both particle masses and spacetime geometry. This is unification in the spirit of Maxwell's: two apparently separate forces turned out to be aspects of one field. Here, an apparently separate force (gravity) turns out to be an emergent aspect of a field already in the theory (Higgs).

"The gravity gap in the unification program is closed: gravity is a QFT phenomenon, the Higgs is the agent, and the four forces are unified in origin."

— *This paper*

SECTION V

What This Framework Resolves

5.1 Why Mass and Gravity Correlate

As argued in §4.3, the perfect correlation between inertial mass and gravitational charge is explained — not assumed. Both are outputs of the Yukawa coupling y_f. The Equivalence Principle is *derived*. This is a significant clarification: GR takes the Equivalence Principle as its foundational postulate without explaining why. This framework answers the why: because both phenomena are produced by the same field coupling.

5.2 Why Gravity Is So Weak

The hierarchy problem — why is gravity so enormously weaker than the other forces? — has a natural answer. The effective coupling of a Standard Model particle to gravity is $y_f v/M_P$ — suppressed by the ratio of the electroweak scale to the Planck scale, $v/M_P \approx 10^{-16}$. The companion paper derives this suppression explicitly. Gravity is weak because it is the spacetime field's response to Higgs coupling, and the spacetime field couples at the Planck scale — seventeen orders of magnitude above the electroweak scale.

5.3 The Axiomatic Basis of General Relativity

General relativity's Einstein field equations provide no mechanism for why mass curves spacetime. The stress-energy tensor is an input; curvature is the output; the causal process is left as an axiom. This framework provides the mechanism. As established in [Author, 2026], the stress-energy tensor is a

derived property of the field interaction structure. The Einstein equations are a theorem — an emergent relationship between geometry and matter that follows from the quantum field dynamics.

5.4 The Incommensurability of QFT and GR

The apparent conflict between quantum field theory and general relativity is resolved in this framework not by quantizing spacetime geometry but by showing that spacetime geometry is an emergent consequence of quantum field dynamics. QFT and GR are not two theories describing the same regime in incompatible ways. They are two levels of description of the same phenomenon: QFT describes the field dynamics that produce the spacetime geometry that GR describes. GR is the classical limit of the Higgs coupling mechanism.

This is analogous to the relationship between thermodynamics and statistical mechanics. Thermodynamics describes macroscopic properties with great precision. Statistical mechanics derives those properties from microscopic dynamics. The two theories are not in conflict — one is the foundation of the other.

SECTION VI

What This Framework Predicts

6.1 Equivalence Principle Violations at Quantum Scales

Different Standard Model particles have different Yukawa couplings y_f. In this framework, their gravitational coupling is mediated by the same Yukawa interaction. The framework predicts small, species-dependent violations of the equivalence principle:

$$\frac{\Delta(m_g/m_i)}{m_g/m_i} \sim \frac{y_f - y_{f'}}{y_f} \cdot \left(\frac{v}{M_P}\right)^2 \sim 10^{-32}$$

Far below current sensitivity but a specific prediction for future atom interferometry [Aguilera et al., 2014; Tino et al., 2020].

6.2 Modified Gravitational Wave Propagation

Gravitational waves propagating through regions of varying Higgs condensate (near the electroweak phase transition) would experience modified group velocity:

$$v_{GW} = c\left[1 + \alpha\frac{|H|^2}{M_P^2}\right]$$

Detectable in principle by LISA [Amaro-Seoane et al., 2017] or pulsar timing arrays via primordial gravitational waves from the electroweak epoch.

6.3 The Brans-Dicke Parameter is Predicted, Not Free

Standard scalar-tensor theories (Brans-Dicke) contain the free parameter ω_{BD}, measured by experiment. In this framework, ω_{BD} is a prediction [Author, 2026]:

$$\omega_{BD} \approx \frac{M_P^2}{8\xi^2 v^2} \approx \frac{1.2 \times 10^{28}}{\xi^2}$$

For $\xi \sim 10^4$: $\omega_{BD} \sim 10^{20}$, vastly exceeding the Cassini bound and explaining why the scalar degree of freedom has not been detected. The prediction is not only consistent with experiment — it predicts the direction of the deviation.

6.4 No New Force Carriers Required

A distinctive prediction, relative to most unification programs: **no new force carriers are predicted at any energy scale**. String theory predicts states near the string scale. GUTs predict X and Y bosons. This framework operates entirely within the Standard Model particle content. As collider energies increase, if no new force carriers appear up to the Planck scale, this framework is confirmed over alternatives that require them.

6.5 LHC Higgs Physics is Gravitational Physics

Because the same coupling ξ appears in Higgs inflation, Higgs self-coupling measurements, and the gravitational sector, precision measurements of Higgs self-coupling at future colliders (FCC-hh, ILC)

Lensing and precision measurements of the gravitational coupling. A convergence of ξ determined independently from Higgs physics and gravitational observations would confirm the Higgs as the gravitational scalar.

SECTION VII

Relation to Other Unification Programs

7.1 Grand Unified Theories (GUTs)

GUTs [Georgi & Glashow, 1974; Fritzsch & Minkowski, 1975] seek to unify the three Standard Model gauge groups under a single larger gauge group at high energy. Proton decay — the paradigmatic GUT prediction — has not been confirmed despite extensive searches (Super-Kamiokande: $\tau_p > 1.6 \times 10^{34}$ years for $p \to e^+\pi^0$ [Miura et al., 2016]). GUTs unify the gauge structure of three forces but leave gravity entirely untouched. The two approaches are complementary: GUT unification of gauge forces at $\sim 10^{16}$ GeV can coexist with Higgs-mediated gravity at all scales.

7.2 String Theory

String theory [Polchinski, 1998] achieves the inclusion of gravity by introducing new fundamental objects at short distances. Significant structural achievements include the derivation of black hole entropy [Strominger & Vafa, 1996] and the AdS/CFT correspondence [Maldacena, 1997]. The non-minimal coupling $\xi|H|^2R$ central to this framework appears naturally in string compactifications [Antoniadis et al., 1994]. String theory may be the UV completion of the mechanism described here: at energies above the string scale, string dynamics govern; below the string scale, the effective QFT description applies.

7.3 Loop Quantum Gravity

LQG [Rovelli, 2004; Thiemann, 2007] quantizes the geometry of spacetime directly, representing it as a discrete network of spin-foam structures. This framework is potentially compatible with LQG as a UV completion: if spacetime geometry at the Planck scale is discrete, the continuous effective metric

modulated by Higgs coupling is an approximation. The Higgs condensate would serve as the medium through which discrete quantum spacetime averages into the smooth classical geometry of GR.

7.4 How This Framework Differs

Program	Unification Mechanism	Gravity?	New Particles?	Status
GUTs	High-energy gauge unification	No	X, Y bosons	Proton decay not observed
String Theory	Extended objects, extra dims.	Yes	Entire string spectrum	Not yet distinguished
LQG	Quantized geometry	Yes	None beyond SM	Lorentz violation (constrained)
Asymptotic Safety	UV fixed point of G_N	Yes	None beyond SM	Higgs mass prediction
This framework	**Higgs coupling → metric**	**Yes**	**None**	**Higgs-gravity correlations, EP violations**

The defining character of this framework: unification **with no new physics beyond the Standard Model**. The Higgs field already exists. The non-minimal coupling $\xi|H|^2R$ is already required by renormalizability. The Yukawa couplings are already known. The unification does not await discovery — it is already in the Lagrangian.

SECTION VIII

Open Questions and Future Work

8.1 Completing the Quantum Description

The companion paper provides a complete description at the classical and semi-classical level. Full quantum treatment requires: (a) one-loop effective action computation verifying induced Newton's constant to leading order; (b) generalization to spin-$\frac{1}{2}$ fermions and spin-1 gauge bosons,

dependence; (b) universality of gravitational coupling; (c) gravitational wave generation rate computation compared to LIGO observations.

8.2 The Cosmological Constant Problem

This framework does not resolve the cosmological constant problem. With $\langle V(H)\rangle_{vac} = \lambda v^4/4 \approx (88\ \text{GeV})^4$, the required cancellation is exact to 55 decimal places — the worst fine-tuning problem in physics. Whether the Higgs-gravity coupling provides a dynamical mechanism for the cosmological constant to relax to its observed value is an open question.

8.3 Measurement of ξ

The non-minimal coupling ξ is the central parameter of this framework, constrained from multiple directions: Higgs inflation requires $\xi \sim 10^4$ [Bezrukov & Shaposhnikov, 2008]; LHC Higgs self-coupling measurements constrain ξ from below; the Cassini bound on ω_{BD} constrains ξ from the gravitational side. A precision measurement combining future collider Higgs data with gravitational wave observations and CMB constraints would provide a critical test.

8.4 Near-Planck Phenomenology

The framework operates classically and semi-classically. At energies approaching $M_P \approx 10^{18}$ GeV, the perturbative treatment breaks down. Whether this framework connects smoothly to a UV completion — string theory, asymptotic safety, or a novel formalism — is one of the deepest open questions.

8.5 The Spacetime Field Ontology

Is the metric $g_{\mu\nu}$ a quantum field — with Fock space, creation and annihilation operators, and a vacuum state — or does it emerge entirely from the collective dynamics of other fields? The core claim of the companion paper stands regardless of the answer. But for a complete theory, the ontological question must eventually be answered.

The unification of the four fundamental forces has been the central ambition of theoretical physics for decades. The ambition was frustrated by gravity — not because gravity was mysterious, but because the obvious approach (quantizing the gravitational field directly) failed.

This paper argues that the frustration was misplaced, because the mechanism for gravity was always in the Standard Model. The Higgs field, through its Yukawa couplings to matter and its non-minimal coupling to spacetime curvature ($\xi|H|^2R$), constitutes a complete, physically motivated, mathematically rigorous mechanism by which particle field excitations produce spacetime geometry. This mechanism was established in the companion paper [Author, 2026].

The unification is the recognition of what that mechanism implies: **all four fundamental forces are now described within quantum field theory**. The strong force is $SU(3)_C$ gauge coupling. The electromagnetic force is $U(1)_{EM}$ gauge coupling. The weak force is $SU(2)_L \times U(1)_Y$ gauge coupling, broken by the Higgs VEV. The gravitational force is Higgs condensate coupling to the spacetime metric.

The Higgs field is the bridge between all four. Mass — the common denominator between electroweak physics and gravity — is Higgs coupling. Gravity is the spacetime response to Higgs coupling. They are not two phenomena that happen to correlate with mass; they are the same phenomenon at different stages of the causal chain.

The framework does not resolve every open problem. The cosmological constant problem, the hierarchy problem, the ontology of the spacetime field, the UV completion at the Planck scale — these remain open. The claim is not that every problem in quantum gravity is solved. The claim is that **the gravity gap in the unification program is closed**: gravity is a QFT phenomenon, the Higgs is the agent, and the four forces are unified in origin.

That is enough to constitute a unified field theory. The rest is detail.

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